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Studierichting: Informatica

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$$\begin{aligned} \int \frac{1}{\cos x - \sin x + 1} dx &\stackrel{t=\tan(\frac{x}{2})}{=} \int \frac{1}{\frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\ &= 2 \int \frac{1}{2-2t} dt = - \int \frac{1}{1-t} d(1-t) = -\ln|1-t| + c \\ &\stackrel{t=\tan(\frac{x}{2})}{=} -\ln\left|1 - \tan\left(\frac{x}{2}\right)\right| + c \end{aligned}$$

References